PRIVOKE: A Client-Side Semantic Privacy Firewall for User Prompts in Web-Hosted LLMs

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***Abstract*—Prompts sent to web-hosted language models can disclose personal and organizational information before a provider's controls apply. To address this exposure, we present PRIVOKE, which inspects supported prompts on the user's workstation and enforces an allow, warn or block decision before transmission. Its update service trains detector candidates from supplied labelled examples, checks version and retention constraints, and streams accepted parameters to clients; independently randomized telemetry remains separate from training. We evaluate this protection and update lifecycle through detection and enforcement measurements, retention checks during updates, and assessments of systems and privacy costs. Updated packages improve agreement on an authored contextual workload, although action failures remain and performance is weak on a separate annotation-presence task, where specificity ranges from 13.36 to 21.76%. In a matched continuation study, aggregate gains do not ensure retention: every full-model candidate fails the retention guard. Installed-browser probes further show bounded request handling, while the cost and telemetry audits reveal limits in measurement and estimation. Together, these results describe an implemented and auditable protection and update lifecycle, while independent contextual ground truth, practical utility and reliable generalization remain open questions.**

***Index Terms*—Prompt privacy, local inference, request enforcement, guarded model updates, local differential privacy.**

I. INTRODUCTION

Web-hosted large language models support writing, coding and analysis [1]–[3], often drawing on contextual information that can also disclose a person's health, a colleague's details or an organization's secrets. Reported workplace disclosures illustrate the risks associated with including such material in prompts [4], [5]. Because transmitted content is already available to the provider, provider-side safeguards address a different point in the disclosure path [6]–[8]. Local intervention therefore offers an opportunity to inspect a supported request before it leaves the browser.

Although local inspection provides a point of intervention, recognizing an entity does not establish whether its disclosure is authorized, and even a correct warning may allow the original content to be transmitted. Similarly, a detector update may improve average accuracy while losing a previously protected case. Monitoring can itself disclose information unless the reporting path has a separate privacy boundary. These distinctions motivate PRIVOKE, a client-side prompt firewall that joins local inspection, explicit request enforcement, guarded detector updates and separately randomized telemetry.

PRIVOKE assigns contextual sensitivity, intended audience and information categories, then maps these findings to ALLOW, WARN or BLOCK. On supported requests, ALLOW and WARN forward the original body, while BLOCK cancels transmission. Since WARN only attempts to display a notice without waiting for acknowledgement, it establishes neither consent nor prevented disclosure. In the configuration studied here, the detector runs locally, while a distinct server-side training path uses supplied labelled examples, validates candidate updates against their exact base, and distributes accepted parameters without rebuilding the extension. User prompts and telemetry do not supply training labels.

The closest work already establishes local prompt protection. Casper combines rules, entity recognition and topic detection in a browser extension [9], [10]; Hide and Seek trains local hiding and recovery models [11]; ProSan trains a local sanitizer that trades privacy hiding against task utility [12]. Building on this prior work, our contribution is an implemented, version-bound detector lifecycle coupled to observable request actions, together with evidence that distinguishes useful gains from regression and rejection. Within this lifecycle, a retention check provides a bounded gate rather than a guarantee of safe adaptation.

We organize the evaluation around three questions:

1) **RQ1: Does PriVoke detect and enforce privacy-sensitive prompt policies effectively?**
2) **RQ2: Can PriVoke update its detector without sacrificing previously protected behavior?**
3) **RQ3: What systems and privacy overhead does PriVoke introduce?**

The answers to these questions remain conditional: authored contextual cases show substantial package gains alongside remaining enforcement harms, while separate annotation-based tasks expose weak specificity. Although the update path works mechanically, aggregate improvement does not ensure retention, and the browser and telemetry measurements establish bounded behavior and costs without demonstrating deployment utility. Keeping these answers together makes the system's contribution and its unresolved gaps explicit.

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GitHub Repository Link

II. Related Work and Contribution

Local protection and sanitization. Casper's three layers directly precede PriVoke's layered inspection [9]. In addition to detecting sensitive content, Casper substitutes identifiers, restores them in responses and offers topic-warning acknowledgement. Its support for custom rules/topics and model replacement also means that it should not be characterized as uniformly static. Hide and Seek (HaS) anonymizes entities before cloud inference and locally recovers them from responses, with trained adversaries testing recovery resistance [11]. ProSan selectively replaces prompt words using task-importance and privacy proxies, then distils a local sanitizer [12]. Together, these systems establish both local processing and training as prior art, while PromptShield studies enterprise role/policy classification [13].

A different measured contract. PriVoke's browser interface acts on requests: BLOCK cancels a supported request, whereas WARN and ALLOW send its original body. Although the detector returns masked spans, these do not rewrite traffic, and the system has no response-recovery stage. Its distinctive engineering focus is instead the guarded path from a candidate tied to an exact model revision through publication to client refresh. To assess this path, we measure paired gains and losses, including those of candidates refused by the guard. The separate telemetry path limits randomized event fields without treating them as training feedback. These are bounded systems contributions, not claims of superior sanitization or uniquely local inference.

Comparison boundaries. Because the reviewed systems evaluate different targets, their published metrics cannot be placed on one privacy-accuracy axis: Casper reports entity/topic detection, HaS reports resistance to whole-text recovery alongside downstream task proxies, and ProSan reports item replacement and automatic utility. We retain version-qualified outcomes and interaction details in Appendix D. Our executed Presidio references use disclosed native entity-recognizer settings [14]; these serve as component references rather than reproductions of Casper, HaS or ProSan and therefore support no matched system superiority claim.

III. System Design and Privacy Boundary

A. Scope and Trust

The system aims to reduce inadvertent personal, third-party or organizational disclosure through supported outgoing prompts, under a trust boundary that includes the workstation, enabled extension, local runtime/supervisor and authenticated model publisher. Compromised clients, malicious publishers, deliberately replaced hooks and provider-held historical information lie outside this boundary. Recognized prompt POST formats through page `fetch` and asynchronous `XMLHttpRequest` are supported; workers, attachments, other transports and changed endpoints need separate validation. Inspection is bypassed when protection is disabled, while optional hosted inference changes the disclosure boundary and is therefore excluded from local-inference measurements.

Figure 1 distinguishes three data paths: prompt inspection remains on the workstation; labelled examples enter a separate training service, from which accepted parameters return to the local detector; and coarse, locally randomized event fields reach a telemetry collector without a path to training. This separation describes what each service receives; it does not establish that every policy decision is correct.

B. Local Inspection and Request Enforcement

A page hook extracts supported prompt text and sends it through the extension background process and a loopback gRPC-Web bridge to a workstation-local Python runtime. A native-messaging supervisor manages the bridge/runtime lifecycle, while local user inference remains separate from the server-side training runtime and does not fall back to it. Figure 2 shows the resulting request contract.

The runtime normalizes text while preserving offsets for original-text evidence. Its default cascade runs regular-expression rules first, followed by enabled named entity recognition (NER) and the semantic classifier. A rule-based BLOCK skips later detectors; other findings are aggregated with the strongest action retained. A failed requested layer blocks an otherwise ALLOW result, while valid non-ALLOW findings remain visible. Although unavailable analysis blocks supported requests, a parameter-stream outage can leave other enabled layers operating. This distinction means that fallback may reduce semantic coverage rather than block every full-pipeline request.

The semantic classifier returns three dimensions: sensitivity (none to highly sensitive), visibility (public to personal-private, or unspecified) and a set of ten information categories, such as health, identity and location. A deterministic policy converts these labels and confidence into an action. Highly sensitive content normally maps to BLOCK and potentially harmful personal information to WARN; identity/location combinations can raise restrictions. Low confidence can moderate the action, but neither these labels nor the confidence score independently establish disclosure authorization or calibrated risk. The exact mapping is given in Appendix A.

We compare historical small classifiers with later packages retaining pretrained BERT encoders/poolers [15]–[18]. These models classify prompts rather than generate conversational responses and are offered in three package sizes, called efficient, balanced and quality. Since their tokenizers, capacities, thresholds and histories differ, the names imply no universal performance ordering and package comparisons cannot isolate pretraining or architecture.

C. Guarded Detector Updates

The update process separates candidate computation from the decision about whether a candidate may replace its base. To support these responsibilities, a training controller constructs bounded labelled batches and held-out guard cases, excluding protected rows/text/groups. A separate server runtime fetches the specified model, executes training and returns model-bound parameter changes and candidate metrics. The updater

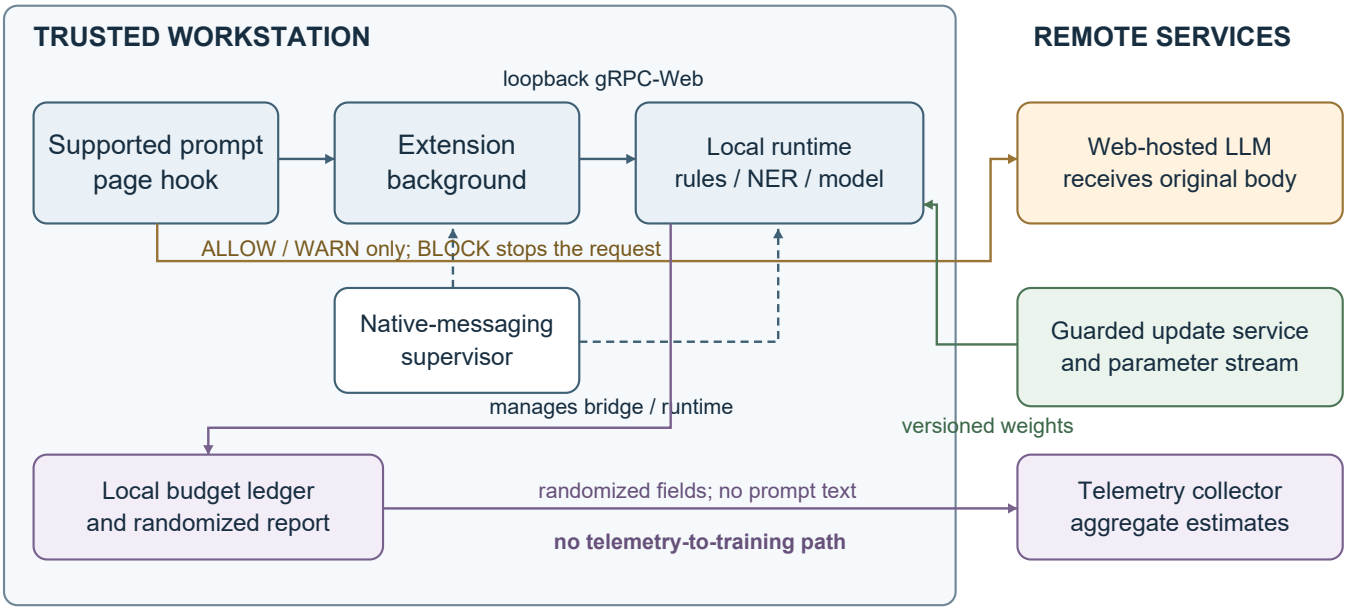


Fig. 1. PriVoke's three data paths. Supported prompt text is inspected inside the trusted workstation. Only an ALLOW or WARN decision releases the original body to the web-hosted LLM. Native messaging manages the local bridge/runtime; analysis uses loopback gRPC-Web. A separate update service supplies versioned weights, and a separate collector receives randomized event fields. Training consumes supplied labelled examples (Figure 3), not intercepted user prompts or telemetry. Dashed lines denote lifecycle control.

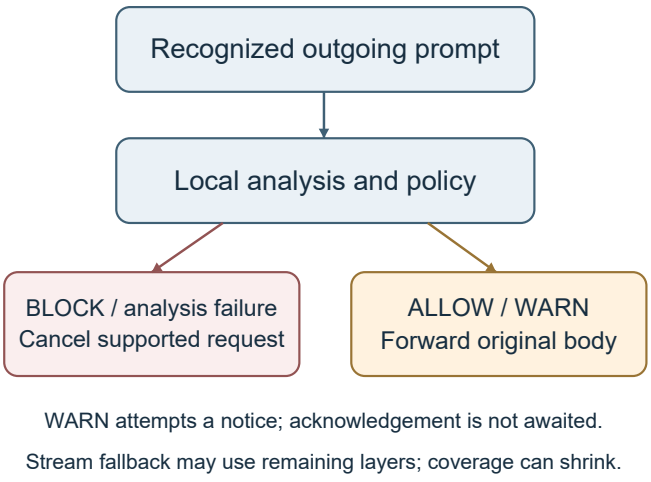


Fig. 2. Pre-transmission enforcement on supported hooks. WARN forwards unchanged content after attempting a notice; it does not wait for user acknowledgement. Analysis failure and reduced-layer stream fallback are distinct conditions. Disabled protection and unsupported traffic lie outside this decision path.

checks exact base identity, allowed trainable tensors and shapes, finite values, magnitude limits and strategy-configured publication guards (Figure 3). Passing an integrity check or successfully receiving an update therefore establishes neither accuracy nor protection.

Validated artifacts are published atomically using compare-and-swap: publication succeeds only if the current base still matches the candidate's declared parent. Durable replay receipts make repeated submissions idempotent for each artifact, while streaming serves versioned parameters whose artifact identities clients verify before refreshing cached weights without an application release. This is centralized training from supplied examples, not federated learning, collection of user gradients or evidence of cloud deployment. This design establishes the ability to update the detector, leaving RQ2 to examine whether particular updates preserve protection.

D. Separate Randomized Telemetry

Telemetry reports five coarse fields: action, an action/sensitivity risk bucket, one precedence-selected category, a bounded release label and a four-hour time-of-day bin without date. Each field is randomized locally before release, and packets exclude raw prompts, identifiers, exact confidence scores and per-layer findings. This reporting path remains separate from training: no telemetry field supplies a corrected label or enters the training path.

For an event budget ε, each of five fields receives $\varepsilon_i = \varepsilon/5$. A field with K_i possible values reports truth with probability

$$p_i = \frac{e^{\varepsilon_i}}{e^{\varepsilon_i} + K_i - 1}, \quad q_i = \frac{1}{e^{\varepsilon_i} + K_i - 1} \tag{1}$$

for each alternative. Under ideal independent sampling and common public configuration, an event tuple has likelihood ratio at most e^{ε}. Because actual finite arithmetic can differ

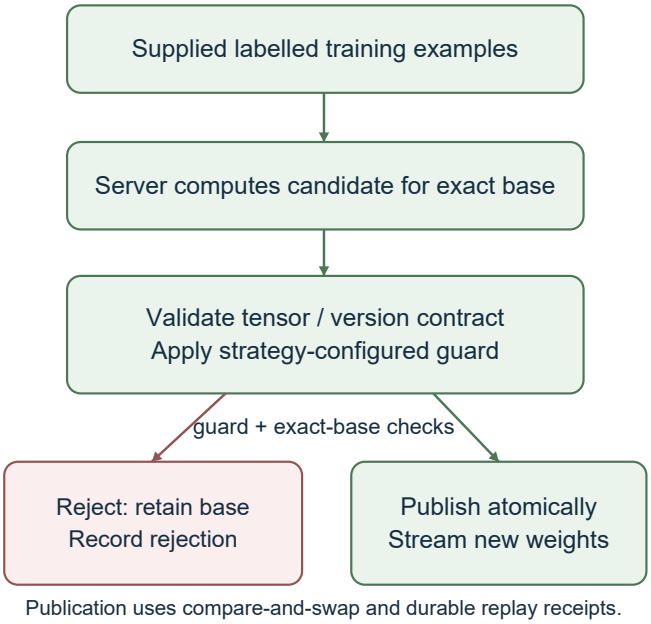


Fig. 3. A candidate is evaluated before publication. Compare-and-swap rejects stale/conflicting bases; durable receipts prevent an identical retry from applying twice. A passing retention guard covers its declared cases and criteria only. Head and underlying-model stages are separate publications, so a later rejection need not undo an earlier accepted head stage.

from this ideal mechanism, it requires a separate audit. A persistent local ledger reserves budget before construction, defaults to event budget one and total eight per UTC day, and does not refund failures. This accounting requires trusted clock/state, while report presence, contribution selection, delivery and network metadata remain outside the content bound. The collector inverts nominal probabilities and clips marginal estimates; randomization can therefore impose substantial estimation error (RQ3).

IV. Evaluation Method

A. Questions and Evidence Units

Table I maps the existing experiments to the three research questions, without counting a reused experiment as independent confirmation when it appears in multiple comparisons. To keep model and product evidence distinct, semantic-only calls explicitly request and verify only the semantic layer, whereas product comparisons request rules, NER and semantics and retain legitimate rule-triggered skips.

Contextual policies. The authored workload contains 1,560 cases: 810 sensitive and 750 clean, with 260 per audience class. Its sensitivity, audience, category and allowed-action targets are assistant-provisional and have not undergone independent human adjudication. For binary scoring, a positive means nonzero sensitivity or any category. Semantic joint correctness requires all three label dimensions; product joint correctness additionally requires an allowed action. An under-restricted action falls below the allowed-action set and an over-restricted action above it. These classification and action outcomes are assessed separately from receiver-observed transmission.

TABLE I
Evidence units and their limits. Presence/category labels are available proxies; independently adjudicated contextual privacy truth is missing.

RQ	Evidence	What it establishes
1	1,560 authored contextual cases; 498 final presence rows; 2,000 external category rows	Agreement with distinct targets; not human privacy accuracy.
1	33 installed-browser receiver probes	Bounded request disposition; not provider coverage or usability.
2	Paired package outcomes; matched head-only/full-model continuation	Casewise gains/harms and guard decisions; not guaranteed adaptation.
3	360 browser decisions in 12 profiles; finite telemetry audit and synthetic simulation	Scoped systems cost and conditional privacy/estimation behavior.

The workload has 35 scenario families and 24 audience-expression groups. Although sealed for an earlier semantic comparison, it was subsequently opened and reused for product and recognizer studies, so it provides PriVoke-authored development evidence rather than independent validation. Repeated passes establish repeatability without increasing the independent sample. Separate paired cluster resampling by family or expression describes dependence sensitivity; it does not certify population coverage.

Annotation-based checks. A once-only final endpoint contains 498 English rows with 236 annotation-positive and 262 annotation-negative targets from PIIMB revision `4a13e9f` [19]. Models, sources and scoring were frozen before release, with no subsequent tuning, reselection or promotion. The resulting prompt-level presence task differs both from contextual authorization and from PIIMB's official character-masking/F2 task. Its 464 observed source-group keys do not certify template independence. A separate external reference uses 2,000 AdvPIIBench rows: 1,000 injected positives, 750 ordinary negatives and 250 hard negatives [20]. Because official source categories are not exhaustive privacy annotations, the selection, grouping and prediction proxies are detailed separately in Appendix B.

B. Comparison and Retention Protocols

Historical and updated native packages are evaluated on the same contextual cases through the product handler, with unchanged rules and policy. Since differences between packages remain confounded, a matched balanced-package ablation provides a narrower contrast by removing individual detectors while fixing texts, weights and policy. Configured small/large English Presidio models provide same-text entity-recognizer references at threshold 0.5 [14], [21], with unsupported contextual/action outputs left unmeasured.

For RQ2, a fixed-base experiment compares head-only and full-model continuation across three seeds. Each treatment starts with byte-identical intact pretrained balanced encoder/pooler and zero task heads, receives the same ordered 4,096 training exposures over 64 steps, and is assessed on the same 1,080 reused provisional cases after complete training/guard group and text exclusions. Full-model continuation alone enables encoder optimization, including the consequent joint-gradient-clipping changes. Assessment includes all terminal candidates, including guard failures, and is not used to select a checkpoint. The 16-case retention guard requires joint improvement and zero action harms against its derived target actions. This guard is smaller and differently scored than the 1,080-case assessment, which permits declared allowed-action sets; neither the weak zero-head control nor the small guard certifies broader safety. Recipes and historical unsuccessful branches remain in Appendices E and B.

C. Enforcement, Cost and Telemetry Protocols

Installed Linux Chromium probes use the built extension, standard native-host installer, native messaging, supervisor, bridge and historical balanced detector. An HTTPS synthetic receiver records original bodies and late sends. Thirty-three correctness probes cover forwarding, blocking, faults, cancellation and recovery, using a synthetic receiver rather than an authenticated LLM provider. Full-pipeline and sole-semantic execution are checked separately, outside the resource windows.

The cost experiment requests rules/NER and measures 360 decisions across twelve fresh profiles, two per transport/action cell. While page-observed latency excludes the receiver response, whole-container CPU and sampled charged memory include resident infrastructure/cache and the observer. These windows provide neither an unprotected baseline nor incremental client attribution, and they do not measure semantic-layer cost. Appendix F specifies boundaries and sampling.

The telemetry audit enumerates finite sampling kernels for one fingerprinted host build, assuming independent uniform OS bits. Separate fixed-contributor simulations quantify estimator error, while real ledger checks exercise persistence, concurrency and failure accounting. The resulting evidence supports conditional mechanism checks and synthetic utility measurements rather than a deployed monitoring study.

V. RQ1: Detection and Enforcement

A. Contextual Policy Agreement

Table II compares historical and updated product packages. All six runs completed without failed calls or layer errors, with updated joint correctness increasing by 27.37–41.86 percentage points. Despite this improvement, exact agreement remained 693–879 of 1,560, and the gains did not preserve every previous success. Paired joint improvements/regressions were 484/57, 573/16 and 671/18 for efficient/balanced/quality. Allowed actions improved on 389/389/348 cases and regressed on 3/6/0. Thus, even quality's zero action regressions coexisted with eighteen joint-label regressions, and no package was promoted.

TABLE II
SAME 1,560 PROVISIONAL CONTEXTUAL CASES, FULL PRODUCT. EACH ENTRY IS HISTORICAL/UPDATED. JOINT AND ACTION COUNTS HAVE DENOMINATOR 1,560; SENSITIVE RECALL COUNTS 810 AND CLEAN SPECIFICITY COUNTS 750. ZERO ERRORS IN EVERY ARM.

Package	Joint	Action	Sensitive	Clean
Efficient	266/693	1029/1415	544/808	494/629
Balanced	160/717	1026/1409	771/802	318/631
Quality	226/879	1073/1421	598/810	468/641

TABLE III
FINAL SEMANTIC ANNOTATION PRESENCE: 236 POSITIVE/262 NEGATIVE ROWS. HISTORICAL/UPDATED COUNTS; TRUE/FALSE POSITIVES (TP/FP), TRUE/FALSE NEGATIVES (TN/FN). ALL SIX ARMS HAVE ZERO ERRORS.

Package	TP	TN	FP	FN
Efficient	140/218	61/45	201/217	96/18
Balanced	141/221	65/35	197/227	95/15
Quality	167/190	51/57	211/205	69/46

The cascade helps explain a recurring burden across the updated packages, all of which over-restricted the same 139 cases: 109 clean controls received rule-based BLOCK and skipped later detectors, while thirty low-risk controls received rule-based WARN. Rules ran on all cases and early-blocked 289; NER and semantics each ran on 1,271. Although loaded NER produced zero converted PriVoke findings after supported-label filtering, this does not imply that it found no raw entities or was inactive.

Removing rules from the balanced package increased allowed-action agreement from 1,409 to 1,530 and joint agreement from 717 to 972. It corrected all 139 shared unnecessary restrictions but lost eighteen previously correct actions; under-restricted cases rose from twelve to thirty. The 121 net action gains therefore coexist with lost protection under the provisional targets. Removing NER left full-product outputs unchanged on this fixture, but neither finding justifies universally disabling a detector. Across the comparison, all 10,920 ablation calls had valid execution records.

B. What Transfers Beyond the Authored Cases?

The separate final annotation-presence endpoint provides a difficult counterpoint to the contextual gains. All 2,988 semantic-only calls completed without errors and verified sole-layer execution, with the confusion counts retained in Table III. Updated recall was 80.51–93.64%, but specificity only 13.36–21.76%. Relative to the historical packages, specificity fell for efficient and balanced, while quality's small increase remained uncertain. Paired net-accuracy gain ranges from frozen source-group resampling were 7.41–17.37, 5.35–15.03 and 1.02–10.71 percentage points, respectively. Average gains therefore do not establish useful clean-label discrimination.

The same updated artifacts achieved 98.40–99.87% specificity on the authored semantic-only contextual fixture. The

contrast holds artifacts and the nonzero-sensitivity-or-category response projection fixed, while changing texts and target labels. The contrast therefore does not represent an intervening model regression, and its causes have not been isolated. Although the mismatch between annotation presence and contextual labels limits interpretation, it does not remove the poor final result, and neither task supplies independently adjudicated contextual privacy truth.

The external source-category check shows a similar tradeoff under another proxy. Balanced updated recall was 92.9% versus historical 63.0%, but source-negative specificity was 0.5% versus 48.6%. This check counts any semantic finding as positive and labels only injected identifier categories, so 0.5% specificity cannot be restated as a proven 99.5% broad-privacy false-alarm rate. It nevertheless indicates poor agreement with the declared source target, while restricted five-type Presidio shows the opposite tradeoff, achieving 99.1% specificity with only 3.9% recall. All 6,000 model/recognizer analyses completed without errors; no threshold or row was changed afterwards.

On the authored contextual cases, small-English Presidio had recall 45.31% and specificity 71.87%, while its larger configuration traded recall down to 29.63% for specificity 88.80%. Both classified 906 of 1,560 correctly, with 192 paired gains and 192 losses between configurations. Updated product packages had recall 99.01–100.00% and specificity 83.87–85.47%, but their entity/contextual task mismatch and PriVoke-authored data preclude a general privacy ranking (Appendix C).

C. Does a Decision Control Transmission?

The complete installed-browser capture observed eighteen unchanged-body sends and fifteen zero-send outcomes across thirty-three declared correctness probes. These outcomes were supported by runtime-call correlation and late receiver checks without a decision broker, while healthy full and sole-semantic probes verified execution. An interrupted parameter stream blocked semantic-only requests; configured full-mode fallback executed rules/NER and forwarded, demonstrating reduced coverage. Restarting restored full execution, whereas disabling protection/layers forwarded requests by design. The completed result is reported separately from nine earlier failed captures, which are not pooled with it.

Answer to RQ1. PriVoke detects many provisional sensitive cases and enforces its declared supported-request contract, including explicit blocking. However, this effectiveness remains conditional, because updated packages still make action errors, WARN transmits original content, independent contextual ground truth is missing, and separate presence/category tasks reveal weak specificity. Receiver probes establish bounded enforcement rather than provider compatibility, prevented disclosure on every sensitive prompt or practical usability.

TABLE IV
MATCHED CONTINUATION ON 1,080 REUSED PROVISIONAL CASES. ENTRIES ARE HEAD-ONLY/FULL-MODEL. JOINT INCLUDES EXACT SENSITIVITY, AUDIENCE, CATEGORIES AND AN ALLOWED ACTION. THE UNCHANGED ZERO-HEAD BASE HAS ZERO JOINT AND 480 ACTION MATCHES. GUARD HARMS USE SIXTEEN SEPARATE CASES.

Seed	Joint correct	Allowed action	Guard harms
42	192/257	908/993	0/1
43	204/247	906/989	1/1
44	198/268	912/1003	0/1

VI. RQ2: Updates and Previously Protected Behavior

A. Mechanism Versus Useful Adaptation

Audited server computation, bounded publication and client refresh demonstrate that served parameters can change, although these mechanics alone do not establish that an update is worth retaining. Earlier contextual, curriculum and accelerated training studies failed their declared qualification criteria. A pretrained package improved provisional disclosure agreement but failed a serious-action veto. These negative results and their attempted budgets are retained in Appendix C, rather than displaced by later package gains.

Later supervised packages improved authored contextual classification, with component and action regressions retained. Because the RQ1 comparison measures the resulting packages rather than automatic online-training benefit, its differences in capacity, tokenizer, threshold and supervision prevent attribution to one update mechanism. The controlled continuation below asks a narrower question with a common base and exposure budget.

B. Matched Continuation and Guard Decisions

All six 64-step fits completed with finite parameters, with head-only training preserving every encoder tensor and full-model training changing 39 encoder tensors while recording nonzero later encoder gradients. Seven snapshots, including the unchanged base, completed 7,560 native in-process semantic assessments without errors. These observations represent repeated assessments of 1,080 situations rather than 7,560 independent cases or deployed network calls.

Full-model continuation improved aggregate joint and action agreement relative to head-only training in all three seeds (Table IV). It nevertheless regressed on fourteen, twenty-six and twenty joint cases, and thirty-eight, forty-one and nineteen actions, respectively. Under-enforcement distance increased on 33/37/15 assessment cases and over-enforcement on 17/13/17; these counts are not a mutually exclusive harm union.

Every full-model candidate failed the fixed guard because of one action harm and would retain the unchanged base. Head-only seeds 42/44 passed, while seed 43 failed. These mappings from guard outcome to retained base are hypothetical, since no candidate was published or promoted. They illustrate why publication decisions must remain separate from average

scores: the guard caught specific harms, but its sixteen cases cannot establish the absence of other harms.

Answer to RQ2. PriVoke supports version-bound updates and can refuse candidates that violate a declared retention test. This support does not establish improvement without sacrificed protection, since matched full-model candidates improved averages yet all failed the guard, while package gains retained casewise regressions. Useful adaptation remains conditional on supervision, representative retention evidence and explicit acceptance criteria.

VII. RQ3: Systems and Privacy Overhead

A. Measured Systems Cost

The completed installed-browser protocol recorded 360 rules/NER decisions in twelve fresh-profile windows. Across transport/action cells, page-observed decision medians ranged from 27.9 to 35.0 ms, with each reported p99 corresponding to that cell's observed maximum. All 240 ALLOW/WARN decisions executed rules and NER, while 120 BLOCK decisions executed rules and legitimately skipped NER. These measurements exclude semantic-layer and receiver-response cost.

Across 126.40 seconds of summed windows, aggregate container CPU was 193.29 CPU-seconds, including 0.69 observer CPU-seconds. Maximum sampled charged memory was 1,267.91 MiB. The container is shared by resident services/cache, the driver and the receiver, while a maximum sampling gap of 622.84 ms leaves open the possibility of missed memory spikes. Although final-read and cleanup checks passed, these scope limitations remain. Without an unprotected baseline or per-role attribution, these totals describe the instrumented system and do not isolate incremental PriVoke overhead. Startup intervals and cell tails appear in Appendix H.

Updated packages had lower observed host-call medians but larger server-process peak RSS than historical packages. Fixed execution order, warming, different backends and cascade shortcuts prevent a controlled speedup or model-only interpretation (Appendix G). Training times and allocated CUDA tensor peaks likewise do not establish comparative training speed or whole-process memory requirements.

B. Privacy Cost and Telemetry Utility

The host finite-kernel audit found that maximum event privacy loss exceeds nominal budgets 0.5, 1 and 2 by approximately $(1.294, 1.503, 1.226) \times 10^{-15}$. The exact nominal bound therefore fails despite the tiny excess, and eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same independent uniform-entropy assumptions. These assumptions leave other builds and deployed entropy uncertified, as discussed in Appendix I, which distinguishes the implemented finite kernel from the ideal formula.

The utility assessment reveals a further cost of randomization: fixed-contributor simulations found substantial clipping bias and root-mean-square error, together with incoherent marginal totals. For eleven-category reports with event budget one and eighty contributors, a true first-category count of seven had clipped bias +22.84 and RMSE 42.32 counts. Marginal estimates failed to sum to eighty in 99.99% of trials, indicating a limitation of the conditional synthetic estimator rather than providing a population monitoring accuracy estimate.

Real ledger checks enforced the daily cap across persistence and concurrent requests, and failed construction/delivery still consumed reservations. Reversing a constructed event sequence changed first-eight BLOCK participation from 100% to 0%. Randomizing report contents therefore does not remove contribution-selection bias or protect whether an event participated.

Answer to RQ3. PriVoke introduces measurable local processing and infrastructure costs, but incremental and semantic-pipeline overhead remain unmeasured. For randomized telemetry, the nominal content bound is conditional and narrowly exceeded, while synthetic estimation error remains substantial. These findings establish neither deployment utility nor protection of participation and network metadata.

VIII. Discussion and Limitations

A. What the System Evidence Supports

The central contribution is a complete protection lifecycle with explicit boundaries: inspect locally, enforce a request decision, evaluate a version-bound candidate before publication, refresh clients, and randomize separate telemetry. The experiments show why those boundaries matter. For example, semantic improvements cannot undo a rule-based early BLOCK, while WARN results still allow exposure. Similarly, a candidate with a larger average gain may be rejected because of a single retained protection failure, and telemetry can support conditional aggregate estimation without becoming a source of training truth.

This contribution complements sanitization and recovery systems rather than replacing their utility questions. Casper, HaS and ProSan motivate what a user may need after detection: an acknowledged intervention, a usable rewritten prompt or a restored answer. PriVoke currently offers a request-action contract, so claims about these additional functions or superiority over the reviewed systems would require separate implementation and evidence.

B. Ground Truth and Generalization Remain Open

The largest scientific gap is independently adjudicated contextual privacy truth. Because our sensitivity, audience, category and action targets are assistant-provisional, they do not establish authorization or human agreement. Source annotation presence and injected category labels provide useful checks without filling this gap, since entity detection, contextual disclosure, appropriate action and task utility require different labels. The large drop from authored clean specificity to final annotation-negative specificity is an observed limitation, not evidence that one target should be dismissed.

The dependence and reuse of the authored cases further limit generalization: final source-group keys and external lexical exclusions do not certify template independence, cover

paraphrases or remove unknown upstream pretraining exposure. Final rows are source-imbalanced: 308 Nemotron, 89 AI4privacy, eighty Gretel, twenty Privy and one MAPA. These English synthetic/machine-generated workloads do not represent other languages or organizational policies, and the separate cluster sensitivities remain descriptive and unadjusted. Neither repeated runs nor three seeds create independent populations. Native-package differences further limit causal attribution. A new protected successor is needed for future tuning after the opened final endpoint.

C. Retention, Availability and Practical Utility

A retention guard can protect only the behavior it tests, leaving considerable uncertainty when the matched guard is small and the zero-head control is weak. Rejected candidates do not become served improvements, and the broader negative studies likewise do not establish useful automatic adaptation. Future studies need common pinned bases, independent retention cases and explicit casewise harm criteria, retaining failed and effective artifacts separately.

Fail-closed behavior also introduces availability costs, while the installed Linux probes cover only the declared historical-detector Chromium/native receiver paths. Authenticated providers, Windows, updated-model installation, workers, attachments and other transports remain unverified; confidence scores are heuristic rather than calibrated joint correctness, and WARN does not wait for informed acknowledgement. No study measures authorized task completion, unnecessary-intervention cost, warning comprehension or reformulation effort, so practical utility and human benefit remain unestablished.

Later instrumentation exposes ephemeral notice-rendering and local request-disposition events. Although fifty-three synthetic host tests validate that contract, its fields postdate the reported browser capture and have not undergone installed-browser or overhead remeasurement. Rendering does not prove comprehension, native handoff does not prove receipt, and page scripts can forge or suppress the events. These events are neither persisted nor sent as telemetry, and measuring utility still requires independent expected actions together with joined user/task outcomes.

D. Measurement and Reproduction Boundaries

The cost measurements have deliberately narrow scopes, since instrumented container totals do not isolate incremental client overhead and rules/NER timings do not measure semantic processing. Similarly, the telemetry bounds depend on retained local state, a trusted clock and independent uniform OS bits. Delivery selection, participation and metadata remain outside these bounds, while synthetic estimator error has no deployment-usefulness threshold and telemetry supplies no corrected contextual labels.

A sealed local source/model capsule reproduced twenty-four invented semantic requests across six packages with matching identities and outcomes. This is bounded replay using existing host dependencies, not public Git-only/fresh-environment reproduction or a protected-final rerun (Appendix I). Ignored pinned model/source assets and licensing constraints remain relevant to redistribution. These limits guide the next work: independent contextual adjudication, protected generalization studies, broader receiver coverage, baseline-relative semantic cost and human utility, rather than additional branches of uncontrolled fitting.

IX. Conclusion

PRIVOKE integrates local prompt inspection, pre-transmission request actions, guarded version-bound detector updates and separate randomized telemetry. Across the three research questions, the evidence yields a qualified systems result. Detection improves on authored cases while specificity transfers poorly, and supported enforcement is observable even though WARN forwards original content. Aggregate gains do not establish useful updates, as illustrated by the rejection of all full-model candidates in the matched study despite their better averages. The measured costs and privacy bounds likewise retain substantial limits in scope and utility. The system provides an auditable mechanism for protection and revision, without established independent contextual accuracy, safe automatic improvement or human utility.

APPENDIX A
EXACT POLICY REPRESENTATION

The sensitivity scale S0–S3 represents no identified risk, low-risk personal context, potentially harmful personal information and highly sensitive/identifying details or secrets. Correspondingly, visibility P0–P4/PU represents public, semi-public, authentication-restricted, group-private, personal-private and unspecified. The ten multilabel categories are HEALTH, POLITICS, RELIGION, CRIMINAL, FINANCIAL, SEXUAL, CHILD, LOCATION, IDENTITY and THIRD_PARTY.

The deterministic policy maps S3 to BLOCK and S2 to WARN. S0/S1 remain ALLOW unless IDENTITY and LOCATION coexist, or either occurs with P2–P4 visibility. Returned confidence below 0.5 moderates S3 to WARN and S2 without IDENTITY/LOCATION to ALLOW; other low-confidence classifications map to ALLOW. Aggregation preserves individual enforcement decisions, unions categories and takes the strongest sensitivity/known visibility. Together, these rules define the implemented policy without independently validating disclosure authorization.

APPENDIX B
DETAILED COMPARISON PROTOCOLS AND TERMINOLOGY

To preserve traceability to the repository records, the protocol identifiers below retain their original names; the historical and updated packages in the main text correspond to V1 and AU3-selected V3. PH3 is frozen-encoder head fitting, EN3 joint continuation and AU3 audience-supervised rehearsal. PIPE3 is the full-product package comparison. HEADS/FULL denote head-only/full-model continuation. These names identify recipes and execution scopes rather than additional research questions.

A. Training Lineage and Selection

Historical online studies bind artifacts, curricula, trainable tensors, seeds, budgets and safety/retention predicates before assessment, retaining accepted and rejected attempts separately. Later paired semantic-only HEADS/FULL and representation studies retain explicit execution scopes, with publication kept distinct from qualification. The repository research records preserve both the full specifications and the unsuccessful attempts.

PH3 fits task heads over frozen pretrained encoders; EN3 jointly continues encoder, pooler and heads; AU3 adds audience supervision with rehearsal. Checkpoint selection uses validation alone, without tuning on the sealed comparison. Since native V1 and V3 differ in capacity, tokenizer, threshold and history, their comparison remains confounded, and the selected task epochs exclude unknown upstream pretraining. Appendix E retains recipes, selected doses and parameter counts.

B. Matched Pretrained Continuation

A separate fixed-base host experiment compares HEADS and FULL continuation for balanced seeds 42/43/44. Each starts from byte-identical intact official pretrained encoder/pooler and zero task heads, distinct from both original V1 and AU3-selected packages. Complete TRAIN/guard group and normalized-text exclusions remove 480 AU3 validation rows before fitting, leaving 1,080 reused provisional cases (570 sensitive/510 clean) in 25 families. Each treatment receives the same ordered 4,096 TRAIN exposures over 64 steps, sensitivity/audience/category objective and dropout-off execution. AdamW head/encoder rates are $10^{-3}/2 \times 10^{-5}$, weight decay zero, clip norm one and microbatch 4/effective batch 64; FULL alone enables encoder gradients and its optimizer group, including consequent joint-gradient-clipping differences. The preceding two-step pilot establishes execution/resource readiness only and is not treated as outcome evidence.

All six terminal candidates and one unchanged ORIGINAL are assessed through native in-process handler requests with explicit successful sole [4] and immutable identities. In this continuation study, joint correctness also requires an allowed action in addition to exact sensitivity, audience and category set; executed layers remain semantic-only. Accordingly, these are 1,080 cases assessed seven times rather than 7,560 independent situations or network RPC evidence, and no assessment is used to select a checkpoint. The maintained 16-case guard requires joint improvement and zero action harms against one derived target action; assessment instead permits its declared allowed-action sets on 1,080 different cases. The mappings between guard candidates and ORIGINAL are hypothetical, since no publication occurs. A bounded guard pass therefore does not certify broader policy safety, particularly when three seeds share one weak zero-head control and the same cases.

C. Product and External-Reference Protocol

PIPE3 compares original V1 and AU3-selected V3 efficient/balanced/quality packages through the native regex-first handler and a narrow inference-only V3 adapter. Requests explicitly select [2,3,4], no audience hints and no presence gate; identities, actual layer status and errors are checked. The comparison uses real installed regex/spaCy detectors, with first-pass same-case accuracy as the primary outcome. Three passes/package yield 28,080 accuracy RPCs plus 24 separate warmups; later identical passes measure repeatability and do not increase independent sample size.

A separate product-scoped ablation fixes the AU3 balanced package, rules, NER, policy and all 1,560 reused cases. Seven explicit configurations request regex [2], NER [3], semantic [4], nonsemantic [2,3], no NER [2,4], no regex [3,4] and full [2,3,4]. Actual execution, legitimate regex skips, identities and errors are checked for every request. This design provides a matched detector-removal comparison for one package and is assessed separately from semantic-only model evaluation. Native V1 windows 64/96/128 and category threshold 0.38 are retained; V3 uses 512 WordPieces and threshold 0.5. Of the three rows exceeding V1 efficient's cap, one regex-skips semantic and two use native truncation; all remain in

the comparison. Windows, tokenizers, capacities, thresholds and histories prevent causal architecture/pretraining attribution. The CPU host uses Python 3.11.9, Torch 2.11.0, NumPy 1.26.4, spaCy 3.7.5 and `en_core_web_sm` 3.7.1 with four model threads. V3 runs before V1 in fixed tier order, one loaded process at a time. Host RPC includes product/transport overhead; peak RSS covers the server process. Because execution order, backends, warming and regex shortcuts differ, the resulting cost measurements remain descriptive.

Independent Presidio 2.2.363 uses the same ordered texts, spaCy 3.7.5, native recognizers and threshold 0.5 [14]. Separate fixed configurations use `en_core_web_sm` 3.7.1 and `en_core_web_lg` 3.7.1 [21]; the larger-model follow-up was frozen before its predictions. Any finding predicts positive, with only text/language entering inference and no target hints, PriVoke rules, mapping, tuning or exclusions. This entity task deliberately differs from contextual binary truth, leaving severity, audience, category exactness, joint and actions unmeasured. Raw same-ID pairing uses existing PIPE3 first passes without rerunning them. The larger-reference freeze omitted an imported analysis-helper hash; independent raw confusion/pairing checks address the reported counts without retrospectively completing that freeze. Given the different execution scopes, direct analyzer timing is not compared with product RPC latency.

Source closures, package/data/recognizer hashes, raw journals and independent audits bind the studies. The tracked compact aggregate/generator reproduce figures without prompts or weights; inference reproduction additionally requires ignored pinned assets, including historical dirty source bytes. A post-hoc raw audit reconciled all 154 declared source files and eighteen model assets to their pinned historical bytes. Relative to clean commit `65a7c78`, 73 source files match exactly, 55 differ only in line endings, eleven differ beyond line endings and fifteen are missing; a byte-exact local archive preserves the required closure. Binary recall, specificity and accuracy use 2,000 paired whole-cluster draws, seed 42 and linear 95% percentile ranges separately by family or audience expression; these describe dependence sensitivity rather than crossed-cluster independence or population coverage. Ten fixed equal-width bins compare returned semantic confidence with observed sensitivity/category exactness where exactly one scored semantic finding was emitted. This outcome-selected subset excludes regex skips and semantic executions without an emitted finding. Since the scalar takes the maximum sensitivity/category-head probability rather than a presence or full-correctness probability, ECE/Brier calibration metrics cannot be inferred from it.

D. External Official-Category Reference

A prospective AdvPIIBench reference [20] freezes revision `02741d9f` and 1,000 injected positives, 750 ordinary negatives and 250 hard negatives. The source is English machine-generated single-turn data without human annotation; only the attacked identifier is labelled, so it does not establish exhaustive five-type or broad privacy-clean negatives. The target is therefore agreement with official CATEGORY, which remains distinct from contextual privacy. Preparation builds whole-source closure over parent IDs, exact/normalized text and typed original injected values before exclusions, then selects globally one row per component, excluding few-shot-safe and protected/conflicting components. All quotas and protected TRAIN/validation/bootstrap exclusions passed before inference; the 2,000 selected components are not certified independent templates.

For this reference alone, any returned semantic finding predicts positive, including WARN; this response proxy differs from the non-S0-or-category projection above. Balanced V1/V3 retain unchanged native settings and use explicit successful sole [4] execution. V1 truncates 477 inputs at its 96-token bound; V3 has no over-cap input at 512, precluding training-only attribution. Presidio lg is separately restricted to credit card, phone, IBAN, email and US SSN at threshold 0.0, unlike the earlier all-recognizer threshold 0.5 reference. Paired component resampling uses 2,000 draws, seed 20261014; one row per component makes it row resampling here. These choices preserve the reference's declared target, but neither finding presence nor CATEGORY agreement measures attacked-span recovery or human privacy accuracy.

APPENDIX C
RETAINED UPDATE AND REFERENCE DETAIL

The unsuccessful branches reported under RQ2 follow separate protocols and are therefore retained individually rather than pooled into a success rate. Early contextual grids completed 126 attempts without a retained qualifier; the curriculum/representation study completed 900 live attempts and 360 offline steps without qualification. None of nine revised realizations in the accelerated 2,016-attempt study met its promising criterion. Disclosure joint correctness remained zero throughout the accelerated 2,016-attempt experiment. The pretrained MiniLM package improved provisional disclosure agreement but failed its serious-action veto. On the sealed AU3 semantic comparison, EN3-to-AU3 joint classification rose from 262/267/429 to 922/972/1209 of 1,560 for efficient/balanced/quality. Paired improvements/regressions were 680/20, 732/27 and 792/12. Clean matches were 738/740/749 of 750. Here, semantic joint excludes action, whereas product joint includes it. This distinction matters because quality retained classification regressions despite no action distance, while efficient/balanced acquired new under-restriction harms. These supervised package outcomes do not establish automatic online-training benefit.

Small-English contextual Presidio confusion was TP/TN/FP/FN 367/539/211/443; large-English was 240/666/84/570, both using native recognizers at threshold 0.5. Updated product efficient/balanced/quality paired binary improvements/regressions versus the small reference were 612/81, 607/80 and 613/68. On a separate historical 502-row development annotation-presence task, the results instead yielded original-product recall/specificity 93.56/19.75% and

configured Presidio 78.79/79.41%; these are different targets and snapshots.

External source-category balanced V1/V3 TP was 630/929 of 1,000; TN 486/5 of 1,000, divided into ordinary TN 321/5 of 750 and hard TN 165/0 of 250. Five-type Presidio returned TP 39, TN 991 (ordinary 750, hard 241). Paired V3-minus-V1 ranges were +26.56–33.16 percentage points recall and -51.13 to -45.21 specificity, with every resample class-defined. Interpretation remains limited by the any-finding proxy and source categories, which do not establish broad-privacy accuracy.

Updated product audience exactness was 1089/1092/1217 of 1,560; category exactness 885/914/979 and category macro F1 .664/.703/.761. Category-bearing joint was 138/149/237 of 720 and multilabel joint 51/69/125 of 456. Correct unspecified-audience labels decreased by 56/65/21 relative to historical packages; efficient unspecified-audience joint fell from 102 to 98 of 260. Across fixed-input repeats, classifications, actions, evidence and traces remained identical.

Post-hoc family resampling gave product binary-accuracy gain ranges 21.64–29.37, 18.28–25.69 and 19.90–29.37 percentage points; expression-group ranges were 23.40–28.08, 19.17–24.81 and 22.44–26.92. These ranges describe separate conditional dependence sensitivities and do not constitute independent generalization evidence.

APPENDIX D
RELATED-SYSTEM INTERACTION DETAIL

TABLE V
INTERACTION FUNCTIONS REPORTED IN THE REVIEWED VERSIONS. "NOT DESCRIBED" IS LIMITED TO THE CITED VERSION AND REVIEW SCOPE.

System	Prompt and response handling	Warning, acknowledgement and request effect
Casper v1 [9]	Replaces identifiers and restores them in responses.	Topic warning has an acknowledgement control; user can inspect or restore.
HaS v1 [11]	Hides entities and locally restores response entities.	Browser warning acknowledgement and request blocking not described.
ProSan v1 [12]	Selectively rewrites prompt words; evaluates downstream utility proxies.	Browser warning acknowledgement and request blocking not described.
PriVoke	ALLOW/WARN forward original body; returned masked spans are not substituted.	WARN notice is not awaited; BLOCK cancels supported requests.

For ProSan's occupation, race and orientation items, the reviewed v1 further requires WordNet similarity below 0.9. Although this criterion constrains the replacements, it does not make the item replacement rate a prompt-level privacy metric.

TABLE VI
EXTERNAL REPORTED OUTCOMES ON DIFFERENT TASKS; NO CROSS-STUDY RANKING OR LOCAL REPRODUCTION.

Work	Selected reported outcome	Population / boundary
Casper v1 (2024) [9]	NER recall 98.5%, specificity 86.7%	Two GPT-4-Turbo-generated 2,000-prompt workloads; NER support is 1,000 positive / 1,000 negative.
Casper 2025 [22]	PII accuracy 92.6%	Published abstract: 4,500 synthetic/2,000 real-world prompts; metric support unavailable. Separate from v1.
HaS v1 (2023) [11]	Black-box label-based protection 0.4070	19,703-example train/test corpus, 80/20; 1.7B black-box attacker; one minus whole-text recovery similarity, not entity protection; exact test support unspecified.
ProSan v1 (2024) [12]	Local-star privacy hiding rate 99.8%	MedQA, SAMSum and CodeAlpaca, 1,000 test prompts each; item-replacement proxy, exact item count unspecified.
PromptShield 2025 [23]	Policy accuracy 72.92%	Independent 435-prompt test; full model/class supports unavailable.

APPENDIX E
SELECTED TRAINING LINEAGE

PH3 fits affine privacy heads over frozen pretrained encoders/poolers using 6,240 TRAIN and 1,560 validation rows, AdamW rate 0.001 and validation-selected checkpoints. EN3 continues each selected parent by jointly fitting encoder/pooler/heads on the same data. AU3 (repository protocol `ba7f3990`) adds 3,120 audience rows to unchanged 6,240-row PH3 rehearsal and uses new 1,560-row validation. EN3/AU3 CPU-four-thread float32 recipes use encoder/pooler rate 2×10^{-5}, head rate 10^{-4}, weight decay 0.01, dropout 0.1 and microbatch 16/effective batch 64. For the final incomplete batch, the loss uses its actual denominator. AU3 warms up 147 updates, allows 5–30 epochs and stops after five validation epochs without strict 10^{-4} loss improvement. Trained checkpoints are selected using validation alone; the sealed comparisons select no model, threshold or recipe.

Table VII gives selected PIPE3 ancestry. Although AU3 completed 16/11/9 epochs, it selected 11/6/4, with discarded work excluded from the selected ancestry. Cumulative selected encoder/pooler epochs are 30/20/8 and heads 42/27/12. V1

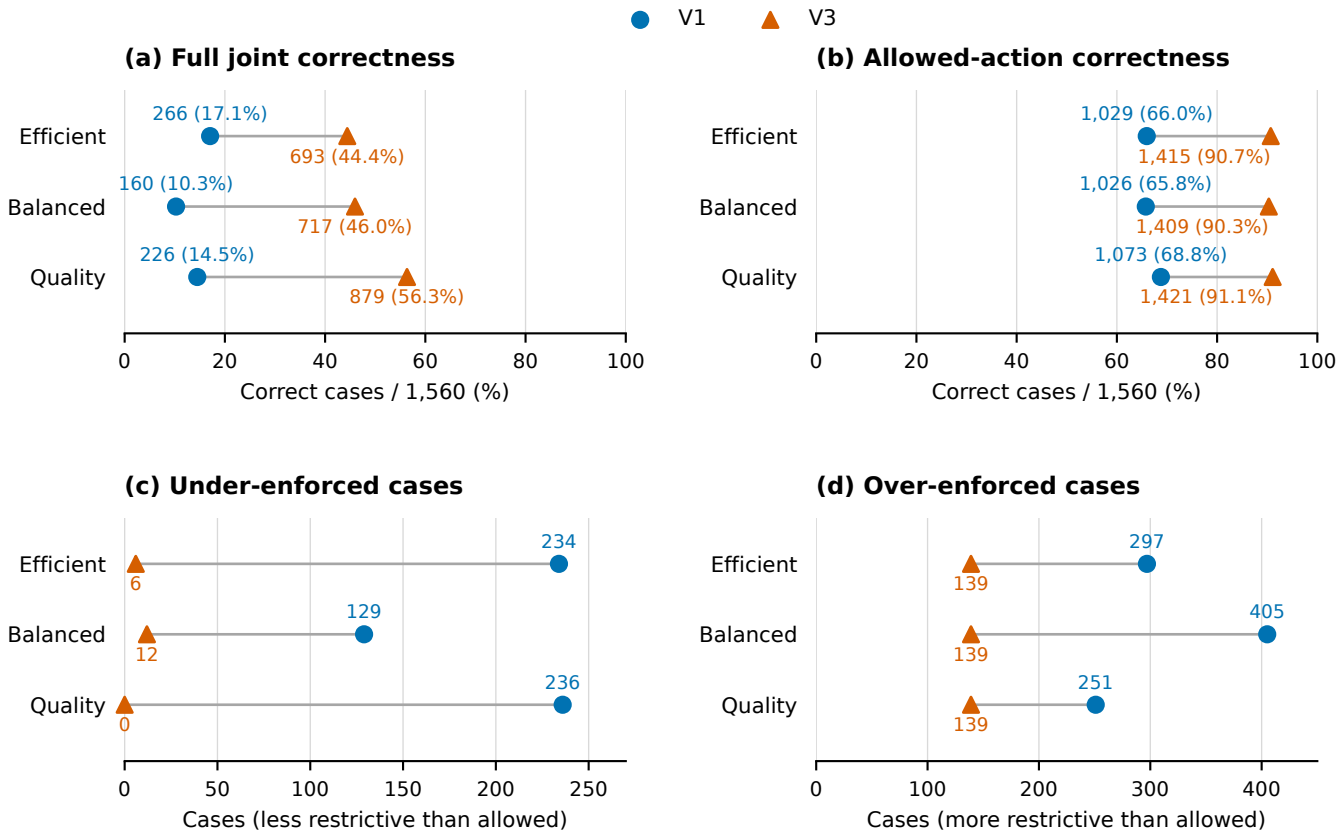


Fig. 4. Detailed product comparison on 1,560 reused provisional cases. V1 denotes historical and V3 updated packages. Under/over-enforced panels count cases on different scales, not ordinal distance; native package differences prevent causal attribution.

encoders have zero optimizer-trained epochs and bootstrap heads 220/360/520. These reported doses concern selected task training and are not equivalent to unknown upstream linguistic-pretraining counts.

TABLE VII
SELECTED V3 ANCESTRY. PARAMETERS ARE V1/V3 MILLIONS; PH3 IS HEAD-ONLY, EN3/AU3 JOINTLY TRAIN ENCODER, POOLER AND HEADS.

Tier	Parameters(M)	PH3	EN3	AU3
Efficient	.019/4.785	12	19	11
Balanced	.037/9.596	7	14	6
Quality	.054/17.494	4	4	4

APPENDIX F
INSTALLED-BROWSER COST PROTOCOL

The completed cost protocol uses twelve fresh profiles, two per transport/action cell, each with one first decision, five warmups and thirty measured requests selecting regex/NER [2,3]; regex BLOCK can skip NER. All thirty-three correctness/fault probes are conducted outside resource windows, using one container that exposes 24 processors without CPU/memory quotas and reuses resident fixture/driver/Go infrastructure. Windows begin before browser launch and end after request/readiness checks, before profile teardown. Thus CPU excludes infrastructure startup before a window, while memory includes resident services/cache. The resulting measurements describe instrumented fresh-profile/native startup and do not include an unprotected baseline.

Decision latency spans page-observed analysis-request to analysis-result messages, excluding receiver response. The pooled nearest-rank percentiles describe two session clusters, with p99 representing the observed maximum rather than an independently estimated tail interval. The collector requests 100-ms intervals and retains every positive actual gap. CPU-counter differences cover first-to-last observations, with a final read beginning after the last request, boundary checks and stop request. Aggregate cgroup CPU includes descendants; sampled charged memory includes cache/kernel charges rather than RSS/PSS [24]. Totals include Chromium, runtime, Go services, fixture, driver and observer, whose own CPU is retained without subtraction. Known browser/supervisor/detector identities, parent links, running status and listeners are checked at boundaries; continuous process attribution/native ancestry and incremental firewall overhead are unmeasured. The protocol also requires cleanup, while the nine failed earlier captures

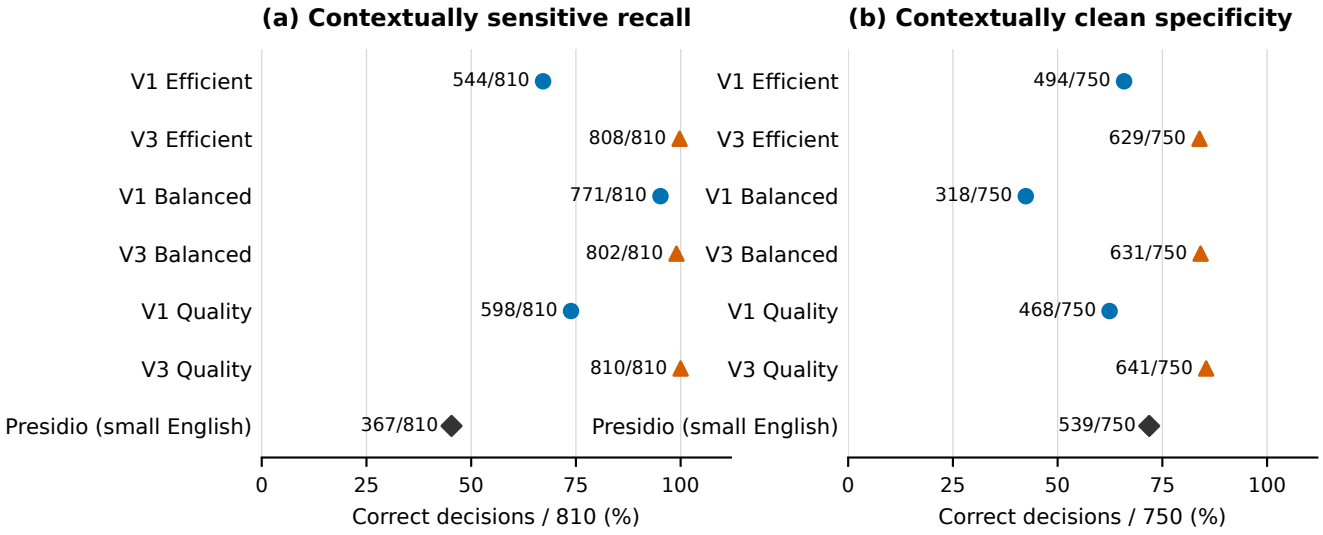


Fig. 5. Same-case binary reference: recall among 810 provisional sensitive and specificity among 750 clean cases. The diamond is small-English Presidio at threshold 0.5. PriVoke-authored reused data and entity/contextual task mismatch preclude an independent privacy ranking.

remain separate and unpooled.

Appendix G
Descriptive Product Cost

Table VIII reports lower observed V3 first-pass host RPC medians alongside larger process peak RSS. Fixed V3-first ordering, different NumPy/Torch backends, warming and cascade shortcuts preclude controlled speedup or model-only cost claims. The historical service timers and Linux lifetime-memory measurements also use different scopes and therefore cannot resolve these attribution limits.

TABLE VIII
PIPE3 DESCRIPTIVE COST: V1/V3 HOST RPC MEDIAN(P95) MS AND SERVER-PROCESS PEAK RSS MIB. EXCLUDES BROWSER OPERATION.

Tier	Median(p95), V1/V3	RSS, V1/V3
Efficient	55.20(70.08)/43.71(50.12)	389.4/466.8
Balanced	65.87(85.65)/42.73(47.82)	389.8/506.4
Quality	59.58(90.08)/50.02(59.48)	391.8/523.4

Appendix H
Installed-Browser Cost Outcomes

The completed protocol yielded twelve fresh-profile windows and 360 measured decisions (Table IX). All 240 ALLOW/WARN measurements executed regex and NER; all 120 BLOCK measurements executed regex and legitimately skipped NER. Nearest-rank p50 values were 1.30 s (range 1.28–9.70) for fresh-profile browser-context creation, 4.37 s (4.21–12.21) for configuration/readiness transition, and 66.4 ms (53.6–75.2) for the first post-readiness decision. These values describe separate instrumented intervals rather than cold-machine or model-only startup.

TABLE IX
INSTRUMENTED PAGE-OBSERVED DECISION LATENCY IN MS FOR THREE FIXED SYNTHETIC CASES, REQUESTED REGEX/NER [2,3]. EACH CELL HAS SIXTY MEASUREMENTS IN TWO FRESH PROFILES; P99 IS THE OBSERVED MAXIMUM. EXCLUDES RECEIVER RESPONSE AND SEMANTIC-LAYER COST.

Transport	Action	p50	p95	p99
fetch	ALLOW	35.0	62.5	73.6
fetch	WARN	31.9	36.1	59.4
fetch	BLOCK	27.9	31.6	33.2
XHR	ALLOW	30.8	35.8	57.7
XHR	WARN	31.3	34.6	59.2
XHR	BLOCK	28.5	32.2	34.1

Across the twelve observed windows (126.40 s summed spans), aggregate container CPU was 193.29 s, including 0.69 s sampled observer CPU. Maximum sampled cgroup charged memory was 1,267.91 MiB, including resident services/cache. The 1,236 samples produced 1,224 within-window gaps: median/p95/maximum 101.21/102.38/622.84 ms, with one over 500 ms. Cumulative CPU and sampled charged-memory maxima differ from wall time and process RSS/PSS, while the sampling gaps can hide memory spikes. Although final-read coverage and process/listener cleanup passed, with natural container exit zero, the whole-container/instrumented scope does not isolate client overhead or support speed comparisons with Table VIII.

Appendix I
Telemetry and Local Reconstruction Details

A fingerprinted Windows CPython 3.11.9 audit enumerated exact rational finite kernels for the actual five domains, $K = 3, 4, 11, 2, 6$. Outward 100-digit logarithm enclosures

place maximum event loss above nominal $\varepsilon = 0.5, 1, 2$ by approximately 1.294×10^{-15}, 1.503×10^{-15} and 1.226×10^{-15}, respectively. Thus the exact nominal bound fails in these settings, despite the tiny excess. Eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same entropy assumptions. These results remain specific to the audited settings, leaving other configurations and deployed builds uncertified.

Across 180 fixed-truth simulation cells and 10,000 repetitions per cell, inversion and clipping produced substantial conditional error. For balanced integer category counts with $K = 11$, event $\varepsilon = 1$ and $n = 80/8000$ contributing reports, first-category true counts were $7/727$, clipped bias $+22.84/+237.50$ and RMSE $42.32/1028.47$ counts. Clipping occurred in $85.62/28.32\%$ of trials; independently estimated marginals failed to sum to n in $99.99/99.78\%$. The larger n pools installation-days, each capped at eight default reservations. These errors concern realized contributor counts and exclude population sampling, while the simulator probabilities are binary64 approximations of the exact finite kernel. Numerical moment/collector checks therefore support implementation consistency without establishing a deployment-usefulness threshold.

Real isolated ledger checks accepted eight reservations and denied the ninth, preserved the cap across process reopen, reset next UTC day, failed closed on rollback/corruption and accepted sixteen of forty concurrent epsilon-0.5 requests. Reservations were also consumed by build, queue and delivery failures. A constructed 32-event sequence with 25% BLOCK yielded 100% versus 0% BLOCK among first-eight contributors when event order reversed. This order sensitivity illustrates that randomization does not remove contribution-selection bias, and no real deployment monitoring accuracy was measured.

A. Local Source/Model Reconstruction

A separately sealed 195-pin local capsule copies the 158 runtime files actually listed in the later FN14 freeze, additional helpers, six frozen V1/AU3 model bundles and prior invented probe receipts. The inventory is assessed separately from the 154-file historical clean-checkout comparison. Six fresh isolated-source host servers replayed the same four invented sole-semantic requests per package: all 24 classifications/actions and four package identity fields matched, with zero errors. Independent post-run checks verified all pins, import origins and process cessation. Together, these checks close a bounded local reconstruction gap using the maintained Python environment, without establishing public Git-only installation, reproduction in a new dependency environment, accuracy or a protected-final rerun.

B. Implemented Finite Kernel

Let $\widetilde{p}_i$ be the binary64 truth threshold. In the audited CPython sampler, fresh uniform 53-bit J selects truth when $J/2^{53} < \widetilde{p}_i$. Hence

$$P_i = \frac{\lceil 2^{53}\widetilde{p}_i \rceil}{2^{53}}, \quad Q_i = \frac{1 - P_i}{K_i - 1}, \qquad \varepsilon_{\text{finite}} = \sum_i \log \max\{P_i/Q_i, Q_i/P_i\}. \tag{2}$$

Alternative selection uses rejection sampling, while the calculation depends on independent uniform OS bits and the fingerprinted host arithmetic. Consequently, it does not certify deployed entropy or other builds.

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